

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 6: Building Our Chemicals of Life Model: Consolidating the Science Behind Two Children in Kisumu

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	C	C	D	C	C

USE AFTER: PREDICT PHASE — Revisiting Lesson 1 prediction cards and writing the Before and After statement

Q1. At the start of the unit, a group wrote: "Food is just energy, so two children eating the same amount should be equally healthy." Which updated statement shows the best evidence-based change in thinking?

- A. Only carbohydrates matter, because they give the body glucose.
- B. Food is made of six chemicals of life with different structures and functions, and missing one can cause a specific disease.**
- C. Food is mainly energy, but children who eat more of it become healthier.
- D. Food with more water in it is healthier because water is a chemical of life.

Correct answer: B. The Before and After shift moves from 'food is just energy' to specific chemical classes, each with its own function and deficiency consequence. The other options keep the idea that quantity or a single nutrient is what matters.

USE AFTER: OBSERVE PHASE — Completing the A3 poster model with the central anchor image of the two Kisumu children

Q2. On a group poster about two children in Kisumu, one child eats only ugali (maize starch) every day and the other eats githeri (maize and beans). Which arrow label correctly links the ugali child to the chemicals of life?

- A. Excess protein and too little carbohydrate
- B. Plenty of carbohydrate but too little protein**
- C. No carbohydrate at all, so no energy
- D. Balanced intake of all six chemicals

Correct answer: B. Ugali is mostly starch, a carbohydrate, so the child gets energy but very little protein. Balanced intake belongs to the githeri child.

USE AFTER: OBSERVE PHASE — Completing the Kenyan Food Sources and lipids part of the poster, recalling the avocado ethanol emulsion result from Lesson 4

Q3. A learner crushes avocado, shakes it with ethanol, then pours the liquid into water. A cloudy white layer forms. What does this ethanol emulsion result show?

- A. Starch is present
- B. Glucose is present
- C. Lipids (fats and oils) are present**
- D. Proteins are present

Correct answer: C. Lipids dissolve in ethanol but not in water, so adding the solution to water forms a cloudy white emulsion. This is the positive test for lipids in avocado.

USE AFTER: EXPLAIN PHASE — Group presentations: why the ugali child develops kwashiorkor

Q4. A child eating only ugali develops a swollen belly (oedema). Which explanation is scientifically correct?

- A. The body has no keratin, so the belly cannot hold its shape.
- B. The child has no DNA to carry the genetic code.
- C. The body cannot make enough albumin, a protein, because too few amino acids come from food.**
- D. Too much starch is stored in the belly as fat.

Correct answer: C. Without dietary protein the body lacks amino acids to make albumin. Low albumin leads to fluid build-up (oedema) and a swollen belly. Keratin is linked to hair discolouration, not the swollen belly.

USE AFTER: EXPLAIN PHASE — Whole-class synthesis and writing the master answer to the driving question

Q5. A child in Kisumu eats githeri (maize and beans) regularly and stays healthy. What is the best chemical explanation?

- A. Beans contain more starch, giving the child more energy than ugali does.
- B. Maize in githeri contains protein instead of starch.
- C. Githeri contains more water than ugali, so the child stays hydrated.
- D. Maize supplies carbohydrate and beans supply protein, providing the amino acids the body needs for protein synthesis.**

Correct answer: D. Githeri combines carbohydrate from maize with protein from beans. The amino acids from the protein let the body build albumin, keratin and muscle proteins, which prevents deficiency.

USE AFTER: DQB PHASE — Driving Question Board review: sorting questions as Answered, Partially Answered or New Question Generated

Q6. During the Driving Question Board review, a learner asks: "How do cells use DNA to know which proteins to build?" How should the class sort this question?

- A. Not a science question, so it should be removed
- B. Answered, because githeri contains protein
- C. New Question Generated, to investigate in a future unit on genetics**
- D. Answered, because amino acids explain kwashiorkor

Correct answer: C. The unit covered that DNA carries the genetic code and that proteins are built from amino acids. How cells read DNA to build specific proteins is a new question that leads into genetics.

USE AFTER: MODEL BUILDING PHASE — Kisumu Menu Evaluation: identifying chemicals that are missing or insufficient

Q7. A typical Kenyan daily menu is: breakfast, uji (thin maize porridge) with sugar; lunch, ugali with sukuma wiki; supper, rice with a small piece of fried fish. Which nutritional gap is most likely?

- A. No carbohydrates, because uji is thin
- B. No lipids, because fried food contains no fats or oils
- C. Insufficient protein, because the only protein source is a small piece of fish**
- D. Too much protein, because fish is eaten

Correct answer: C. Uji, sugar, ugali and rice are mostly carbohydrate, and sukuma wiki adds vitamins and minerals. The small piece of fish is the only protein source, so protein is insufficient. Frying adds lipids, and the menu has plenty of carbohydrate.